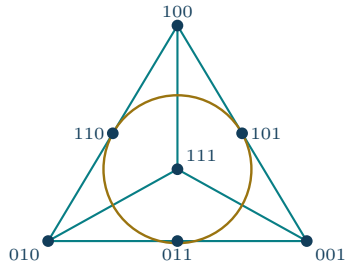
PROJECTIVE GEOMETRY LADDER 9 OF 9

Finite fields, the Fano plane and the whole picture

Over a field with p elements the plane has p² + p + 1 points, and the missing roots show why the field must grow.

1 / THE RULE AND THE PICTURE

Seven points and seven lines, every two lines meeting in exactly one point



the Fano plane, with points as binary vectors

Using $\mathbb{F}_2 = \{0, 1\}$ in homogeneous coordinates, the points are the seven non-zero vectors of $\mathbb{F}_2^3$ and a line is a set of three that add to zero, such as $110, 011, 101$, the circle in the picture. Every two points lie on one line and every two lines meet in one point, and in general

$$\boxed{|\mathbb{P}^2(\mathbb{F}_p)| = p^2 + p + 1}$$

with $p + 1$ points on each line and $p + 1$ lines through each point.

2 / THE SAME ALGEBRA MOD p

Cross products join and meet in any field

Over $\mathbb{F}_3$ the lines $x + y = 1$ and $x + y = 0$ are $[1 : 1 : -1]$ and $[1 : 1 : 0]$ and their cross product is $(1 : -1 : 0) = (1 : 2 : 0)$, a point at infinity, so parallel lines meet exactly as over the reals.
A line has $p + 1$ points, the p ordinary ones and one at infinity, and the whole plane has p^2 ordinary points together with $p + 1$ at infinity, which totals $p^2 + p + 1$.

3 / THE FIELD MATTERS

The circular points exist only when -1 is a square

Setting $Z = 0$ in the circle $X^2 + Y^2 = Z^2$ gives $X^2 + Y^2 = 0$, which has the roots $(1 : \pm i : 0)$ only if $i^2 = -1$ has a solution in the field. Modulo 5 it does, with $i = 2$, and modulo 13 with $i = 5$, but modulo $3, 7, 11$ it does not, so the two circular points exist only in the larger field with p^2 elements, which is how Bézout's count needs an algebraically closed field.

4 / COUNTING THE CIRCLE

Always $p + 1$ points, as many as the projective line

p	points of the plane	affine circle points	circular points	all circle points	$p \bmod 4$
3	13	4	0	4	3
5	31	4	2	6	1
7	57	8	0	8	3
11	133	12	0	12	3
13	183	12	2	14	1

The circle has $p - 1$ affine points when $p \equiv 1 \pmod 4$, with two at infinity, and $p + 1$ affine points when $p \equiv 3 \pmod 4$, with none at infinity, so the total is $p + 1$ either way, which is the parametrisation of the Pythagorean sheet working modulo p.

5 / WHAT EACH INGREDIENT REPAIRS

The ladder in one table

ingredient	what it repairs	example
points at infinity	parallel lines	$x + y = 1$ and $x + y = 3$ meet at $(-1 : 1 : 0)$
an algebraically closed field	missing roots	$y = -1$ meets $y = x^2$ only at $x = \pm i$
multiplicity	tangency	$y = 0$ touches $y = x^2$ twice at the origin
homogeneous coordinates	a separate rule for infinity	one cross product for every pair of lines
no common component	infinitely many points	two copies of one line

6 / TRY THESE

Count and check

(a) How many points and lines has $\mathbb{P}^2(\mathbb{F}_3)$? $9 + 3 + 1 = 13$ of each.
(b) Check that $(1 : 2 : 0)$ lies on $X^2 + Y^2 = 0$ over $\mathbb{F}_5$: $1 + 4 = 5 \equiv 0$.
(c) Why does the Fano plane have no parallel lines? Any two of its seven lines share a point, as the table of lines shows.

ASIDE / THE WHOLE LADDER

Points at infinity, homogeneous coordinates, an algebraically closed field and multiplicity together turn "usually mn" into "always mn".

LINKS TO OTHER SHEETS

The ladder runs: 1 point at infinity, 2 homogeneous coordinates, 3 homogenising curves, 4 Bézout, 5 projective maps, 6 Pappus and Desargues, 7 Pythagorean triples, 8 cubics, and this one; primes modulo 4 are in the number theory ladder.

Convention. $\mathbb{F}_p$ is the field with p elements; $\mathbb{P}^2(\mathbb{F}_p)$ its projective plane; the Fano plane is $\mathbb{P}^2(\mathbb{F}_2)$; points are non-zero vectors up to scale.

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